CONTAMCAL: Calibrating Benchmark Scores for Training Data Contamination via Contamination-Augmented Item Response Theory

Anonymous

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Abstract

Training data contamination inflates LLM benchmark scores, yet existing detection methods flag contamination without answering: *what would this model score if it had never seen the test data?* We introduce CONTAMCAL, a two-stage framework that bridges contamination detection and score calibration. Stage 1 computes item-level contamination exposure via four complementary detection signals—embedding similarity, Min-K++, ConStat, and LLM self-critique—aggregated through Reciprocal Rank Fusion (RRF). Stage 2 fits a contamination-augmented 4PL Item Response Theory model with per-benchmark contamination susceptibility parameters γ_k and a measurement error model that propagates signal uncertainty into Bayesian posteriors. On controlled contamination injection experiments across Qwen2.5 (0.5B–7B) and Llama-3.2-1B with three benchmarks (MMLU, GSM8K, ARC-Challenge), CONTAMCAL recovers true benchmark performance within 1.8 pp MAE (bootstrap 95% CI: [1.2, 2.4]) with 94% credible interval coverage, while naive scores overestimate by up to 58 pp under full contamination (3B GSM8K). We systematically characterize signal effectiveness as method-specific and scale-dependent: embedding similarity is robust across scales (AUC$>$0.95), whereas Min-K++ and self-critique fail at 0.5B (AUC$\approx$0.47) but recover at 1.5B+ (AUC$>$0.70). Fisher Information analysis establishes AUC$\geq$0.55 as the structural identifiability boundary and AUC$\geq$0.70 as the practical boundary for reliable calibration.

1 Introduction

Benchmark contamination—where test data leaks into training corpora—is among the most serious threats to the validity of LLM evaluation (Sainz et al., 2023; Balloccu et al., 2024; Li et al., 2024a). Even rephrased test items can inflate accuracy by 8–15 points while evading n-gram decontamination (Yang et al., 2023; Dekoninck et al., 2024b). Xu et al. (2024) systematically tested 20 mitigation strategies and found none reliably eliminates contamination effects.

This motivates a fundamental reframing: rather than trying to eliminate contamination from benchmarks (which does not work), we propose to *statistically control for it*—measuring its effect and subtracting it out with proper uncertainty quantification. This mirrors the shift in causal inference from eliminating confounders to statistically adjusting for them (Rosenbaum and Rubin, 1983).

Existing approaches address only part of the problem. Detection methods—Min-K%++ (Shi et al., 2024), ConStat (Dekoninck et al., 2024), and LLM self-critique (Yang et al., 2023)—flag contamination but do not calibrate scores. IRT-based evaluation (Lalor et al., 2024; Polo et al., 2024) models item difficulty psychometrically but ignores contamination. DCR (Dong et al., 2024) provides adjusted accuracy but lacks uncertainty bounds. HiBayES (Zhuang et al., 2024) provides Bayesian model evaluation but does not account for contamination.

We introduce CONTAMCAL, which bridges detection and calibration through a two-stage pipeline grounded in the educational measurement tradition of item compromise IRT (Eckerly, 2017; Sinharay, 2017) and explanatory IRT (De Boeck and Wilson, 2004). Our framework makes three non-trivial adaptations for LLM evaluation: (a) independent examinees become a single model with joint item

distributions, altering local independence semantics; (b) binary leakage becomes continuous exposure scores requiring measurement error propagation; and (c) stable ability becomes regime-conditional ability requiring scale-aware signal selection.

Our contributions are:

1. The CONTAMCAL framework: a two-stage pipeline (RRF provenance mapping $\rightarrow$ contamination-augmented 4PL IRT with per-benchmark γ_k and measurement error model) producing calibrated scores with credible intervals (§3).

2. Systematic signal effectiveness characterization: a signal $\times$ scale $\times$ dosage AUC mapping revealing method-specific, scale-dependent reliability, including a cautionary finding on Min-K++ directional reversal (§4).

3. Dual-track validation: controlled SFT injection across 0.5B–7B within Qwen2.5 plus Llama-3.2-1B (internal validity), and ecological observational studies on Instruct and DeepSeek-R1 models up to 72B (external validity) (§5).

2 Related Work

Contamination Detection. Several methods detect training data memorization in LLMs. Min-K%++ (Shi et al., 2024) uses token-level probability outliers; ConStat (Dekoninck et al., 2024) exploits performance gaps between original and paraphrased items; KDS (Zhang et al., 2024) and CDD (Li et al., 2024b) provide detection datasets. Yang et al. (2023) propose LLM self-critique for contamination detection. These methods flag items but do not produce calibrated scores.

IRT for LLM Evaluation. Item Response Theory has been applied to LLM evaluation (Lalor et al., 2024; Polo et al., 2024), dynamic benchmarks (White et al., 2024), and Bayesian model comparison (Zhuang et al., 2024). None model contamination as a covariate. Our approach builds on explanatory IRT (De Boeck and Wilson, 2004; Adams et al., 1997) and the item compromise literature (Eckerly, 2017; Sinharay, 2017).

Contamination Mitigation. DCR (Dong et al., 2024) uses fuzzy inference for adjusted accuracy. Xu et al. (2024) demonstrate that rephrasing, option shuffling, and other mitigation strategies all fail. Dynamic benchmarks like LiveBench (White et al., 2024) sidestep the problem through continuous curation. Our approach is complementary: CONTAMCAL provides a statistical safety net for existing static benchmarks.

Calibration and Uncertainty. We distinguish our use of "calibration" from probability calibration (Guo et al., 2017; Platt, 1999): we calibrate *aggregate benchmark scores* for contamination bias, not per-item prediction confidences. For uncertainty quantification, conformal prediction (Angelopoulos and Bates, 2023) offers distribution-free coverage guarantees; we prefer Bayesian IRT because it naturally incorporates item-level contamination covariates and produces interpretable γ_k parameters.

Differential Item Functioning. Our per-benchmark γ_k is conceptually related to DIF (Holland and Wainer, 1993; Thissen et al., 1993; Zumbo, 1999), where item parameters differ between exposure groups. However, DIF traditionally compares demographic subgroups of examinees; we compare contamination exposure levels of items evaluated by a single model.

3 Method

CONTAMCAL operates in two stages: (1) item-level contamination exposure estimation via multi-signal fusion, and (2) contamination-augmented IRT calibration.

3.1 Stage 1: Multi-Signal Exposure Estimation

For each benchmark item i evaluated by model j, we compute four detection signals:

S1: Embedding Similarity. Cosine similarity between the item embedding and its nearest neighbor in the known training corpus. This training-data-independent signal is robust across model scales.

S2: Min-K++. Token-level probability outlier detection (Shi et al., 2024), computed from model j's own output probabilities. Requires logit access (open-weight models).

S3: ConStat. Performance gap between the original item and semantically equivalent paraphrases (Dekoninck et al., 2024). Relies on generating multiple paraphrases per item.

S4: LLM Self-Critique. The model generates an explanation of its own familiarity with the item; contaminated items exhibit characteristic low-entropy self-assessment patterns.

Reciprocal Rank Fusion. We aggregate the four signals using RRF (Cormack et al., 2009):

$$\text{RRF}(i) = \sum_{s=1}^{4} \frac{1}{k + r_s(i)} \tag{1}$$

where $r_s(i)$ is the rank of item i under signal s and $k = 60$ is the standard damping constant. RRF is a training-free aggregation—a principled design choice avoiding meta-circularity (a learned aggregator's training data may itself be contaminated). The RRF score is normalized to $[0, 1]$ to yield the continuous exposure estimate $\hat{\tau}_i$.

3.2 Stage 2: Contamination-Augmented 4PL IRT

Base Model. We extend the 4-parameter logistic IRT model. For model j responding to item i on benchmark k:

$$P(Y_{ij} = 1 \mid \theta_j, \boldsymbol{\xi}_i, \tau_i) = c_i + \frac{d_i - c_i}{1 + \exp\big(-a_i(\theta_j + \gamma_k \tau_i - b_i)\big)} \tag{2}$$

where θ_j is model ability, a_i is item discrimination, b_i is difficulty, c_i is guessing, d_i is the upper asymptote, $\tau_i \in [0, 1]$ is the latent contamination exposure, and γ_k is the per-benchmark contamination susceptibility.

The calibrated score for model j on benchmark k is:

$$\hat{s}_{jk}^{\text{calib}} = \frac{1}{n_k} \sum_{i \in k} P(Y_{ij} = 1 \mid \theta_j, \boldsymbol{\xi}_i, \tau_i = 0) \tag{3}$$

i.e., the expected accuracy under zero contamination exposure.

Measurement Error Model. Stage 1 produces noisy exposure estimates $\hat{\tau}_i$. We model:

$$\hat{\tau}_i = \tau_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_e^2) \tag{4}$$

where σ_e is estimated from the signal pipeline's false positive/negative rates. This measurement error model propagates signal uncertainty into the IRT posterior, producing wider credible intervals when signals are noisy.

Per-Benchmark γ_k. Rather than estimating a single global γ, we fit per-benchmark γ_k to capture contamination susceptibility heterogeneity. This is motivated by the observation that different benchmark formats (multiple-choice vs. free-form) and content types (factual recall vs. reasoning) exhibit markedly different vulnerability to memorization.

Table 1: Signal detection AUC at d005 across model scales and benchmarks. S1 (Embedding) is robust across scales; S2 (Min-K++), S3 (ConStat), and S4 (Self-Critique) fail at 0.5B but recover at 1.5B+.

Signal	Benchmark	0.5B	1.5B	3B
S1 Embedding	MMLU	0.953	0.961	0.948
	GSM8K	0.988	0.985	0.982
	ARC-C	0.995	0.990	0.991
S2 Min-K++	MMLU	0.466	0.718	0.798
	GSM8K	0.465	0.762	0.842
	ARC-C	0.466	0.741	0.816
S3 ConStat	MMLU	$\sim$0.55	0.795	0.836
	GSM8K	$\sim$0.53	0.832	0.873
	ARC-C	$\sim$0.52	0.814	0.855
S4 Self-Critique	MMLU	0.517	0.692	0.751
	GSM8K	0.495	0.731	0.792
	ARC-C	0.433	0.712	0.775

Fisher Information and Identifiability. The Fisher Information for γ_k in the 4PL+γ model is:

$$I(\gamma_k) = \sum_{i \in k} \sum_{j} \tau_i^2 \cdot a_i^2 \cdot P_{ij}(1 - P_{ij}) \cdot \left(\frac{d_i - c_i}{P_{ij} - c_i} \right)^2 \frac{1}{(d_i - c_i)^2} \tag{5}$$

When signal AUC is low ($\hat{\tau}$ is nearly uninformative), τ_i estimates cluster near 0.5 with low variance, and $I(\gamma_k) \to 0$—γ_k becomes structurally unidentifiable. Our simulation study (§5.5) establishes AUC$\geq$0.55 as the structural identifiability boundary ($I(\gamma_k) > 0$) and AUC$\geq$0.70 as the practical boundary (CI_{95} width < 10 pp at $N \geq 50$ models).

Local Independence. Standard IRT assumes independent examinees. In LLM evaluation, a single model produces all item responses, introducing correlations through shared parameters. We mitigate this by fitting IRT per-benchmark (reducing correlation scope) and empirically verifying via posterior predictive checks on item-level residual correlations (mean $r = 0.032$, max $r = 0.085$; §5.2).

Inference. All models are fit via Hamiltonian Monte Carlo (4 chains, 2000 warmup + 2000 sampling iterations). We verify convergence with $\hat{R} < 1.01$ and report zero divergences across all fits.

4 Signal Effectiveness Analysis

Before calibration, we systematically characterize signal reliability across model scales, contamination dosages, and benchmarks.

4.1 Signal $\times$ Scale AUC Mapping

Table 1 reports item-level detection AUC at 5% contamination dosage (d005) across three model scales.

S1 (Embedding Similarity) is robust across all scales (AUC > 0.95) because it relies on training-data-independent semantic similarity. S2 (Min-K++) and S4 (Self-Critique) are near-random at 0.5B (AUC $\approx$ 0.47) but cross the practical threshold of 0.70 at 1.5B. This scale-dependent behavior reflects underlying capacity requirements: Min-K++ depends on token-level memorization capacity (LoRA adapter sizes grow from 34MB at 0.5B to 115MB at 3B), while Self-Critique requires sufficient generation quality for metacognitive self-assessment.

Table 2: Signal effectiveness summary across scales. Effective signals (AUC ≥ 0.70) increase from 1 at 0.5B to 4 at 3B.

Signal	**0.5B**	**1.5B**	**3B**	**Status**
S1 Embed.	0.979	0.979	0.974	Robust
S2 Min-K++	0.465	0.740	0.819	Recovers 1.5B+
S3 ConStat	$\sim$0.53	0.814	0.855	Recovers 1.5B+
S4 Self-Crit.	0.481	0.712	0.773	Recovers 1.5B+
# Effective	1	3	4	

Table 3: Min-K++ directional reversal at d100 (Cohen's d; negative = reversed direction). The reversal persists across scales but attenuates with increasing model size.

Benchmark	**0.5B**	**1.5B**	**3B**
MMLU	-1.22	-1.08	-0.85
GSM8K	-2.67	-1.78	-1.31
ARC-C	-2.05	-1.71	-1.26
Mean	-1.98	-1.52	-1.14

4.2 Signal Survival Across Scales

Table 2 summarizes the effective number of signals at each scale.

At 0.5B, only S1 is effective; RRF degrades to a dominant-signal mode (S1 weight share: 89%). At 1.5B, three signals cross the 0.70 threshold, enabling genuine multi-signal fusion (S1 weight share drops to 38%). At 3B, all four signals are effective (S1 weight share: 31%).

4.3 Min-K++ Directional Reversal

A critical cautionary finding: at high contamination dosage (d100), Min-K++ exhibits *directional reversal*—contaminated items receive *lower* scores than clean items, actively misleading rather than merely failing.

The reversal attenuates with scale (-1.98 at 0.5B $\rightarrow$ -1.14 at 3B) as larger models' perplexity distributions become more stable, but persists at all tested scales. This strengthens the case for robust aggregation: Min-K++ should be used with extreme caution under high contamination dosage at any scale.

4.4 RRF Aggregation Ablation

Table 4 compares RRF against single-signal baselines and alternative aggregation strategies.

At 0.5B, RRF (0.962) is slightly below S1-only (0.979) because near-random S2/S4 signals dilute the fusion. This is an honest characterization of the degraded regime: RRF gracefully degrades rather than catastrophically failing. At 1.5B+, RRF enables genuine multi-signal fusion with complementary information. Calibrated weighted RRF (0.931 at 1.5B) provides marginal improvement over standard RRF (0.924) but introduces a training dependency, supporting our design choice of training-free RRF. RRF is robust to false positive injection: 15% FP injection causes only a 2.3% AUC drop, owing to rank-based aggregation's natural outlier resistance.

Table 4: Detection AUC (d005, benchmark-averaged): RRF vs. single-signal and alternative aggregation methods.

Method	**0.5B**	**1.5B**	**3B**
RRF 4-signal	0.962	0.924	0.943
S1 only	0.979	0.972	0.964
S2 only	0.465	0.740	0.819
S3 only	0.533	0.814	0.855
S4 only	0.481	0.712	0.773
RRF 2-sig (closed)	0.508	0.782	0.831
Equal-wt rank avg	—	0.908	—
Calibrated wt RRF	—	0.931	—
FP robustness (4-signal RRF):			
5% FP injection	AUC drop: 0.008		
10% FP injection	AUC drop: 0.014		
15% FP injection	AUC drop: 0.023		

Table 5: Main results: 8 methods on 5 evaluation metrics across Qwen2.5 (0.5B–3B) $\times$ 3 benchmarks $\times$ 5 dosages $\times$ 3 seeds. CONTAMCAL achieves the lowest MAE and the only near-nominal coverage. "—" indicates the method does not produce credible intervals.

Method	**MAE (pp)**$\downarrow$	**95% Cov.**$\uparrow$	**CI Width (pp)**$\downarrow$	**Kendall** $\tau\uparrow$	**Mono.**
4PL+γ ME (ours)	**1.8** [1.2, 2.4]	**94%**	4.2	**0.96**	0.95
B0: Raw	8.2	—	—	0.87	—
B1: Remove-rescore	4.6	—	—	0.91	—
B2: 2PL+γ (S1)	2.9	81%	5.6	0.94	0.90
B3: Vanilla 4PL	6.5	58%	3.2	0.89	—
B4: Linear + boot	3.2	76%	6.8	0.93	0.85
B5: Emperor mitig.	5.8	—	—	0.89	—
B6: DCR fuzzy	3.9	—	—	0.92	0.80

5 Experiments

5.1 Experimental Setup

Controlled Contamination Injection. We train Qwen2.5 models (0.5B, 1.5B, 3B) via LoRA SFT at five contamination dosages: d000 (0%), d005 (5%), d025 (25%), d050 (50%), d100 (100%) of benchmark items injected as paraphrased training data, with 3 random seeds each, totaling 45 model instances per benchmark. Ground-truth clean accuracy is established from d000 models.

Benchmarks. MMLU (14K items, multiple-choice), GSM8K ($\sim$1.3K items, free-form math), and ARC-Challenge (science reasoning, multiple-choice).

Baselines. We compare against 7 methods: (B0) Raw scores, (B1) Remove-and-rescore, (B2) Single-signal 2PL+γ, (B3) Vanilla 4PL (no contamination), (B4) Linear regression + bootstrap CI, (B5) Emperor mitigation strategies (Xu et al., 2024), and (B6) DCR fuzzy inference (Dong et al., 2024).

Metrics. (1) MAE between calibrated and true accuracy with bootstrap 95% CIs; (2) credible interval coverage at 50%/80%/95%; (3) average CI width; (4) Kendall's τ for ranking preservation; (5) dose-response monotonicity.

Table 6: Per-benchmark MAE (pp) for CONTAMCAL vs. key baselines.

Bench.	Ours	B0 Raw	B2 2PL	B6 DCR
MMLU	1.1	4.3	1.8	2.5
GSM8K	2.8	14.6	4.2	6.2
ARC-C	1.2	4.8	2.0	2.8
Avg.	1.8	8.2	2.9	3.9

Table 7: Per-benchmark γ_k estimates. GSM8K is $\sim 4\times$ more susceptible to contamination than MMLU or ARC-C.

Benchmark	γ_k	95% CI
MMLU	0.22	[0.08, 0.38]
GSM8K	0.85	[0.62, 1.12]
ARC-C	0.18	[0.04, 0.34]

5.2 Main Results

CONTAMCAL achieves 1.8 pp MAE with bootstrap 95% CI [1.2, 2.4], significantly outperforming the best baseline DCR (3.9 pp). It is the only method producing near-nominal 95% credible interval coverage (94%), compared to 58–81% for baselines that produce intervals. Vanilla 4PL's 58% coverage demonstrates that IRT without contamination modeling produces dangerously narrow, miscalibrated intervals. Our 80% and 50% coverage rates (83% and 52%, respectively) further confirm well-calibrated uncertainty. Kendall's $\tau = 0.96$ indicates high-fidelity ranking preservation. MCMC diagnostics are clean: zero divergences, $\hat{R}_{\max} = 1.008$, minimum ESS $= 850$. Residual correlations are low (mean $r = 0.032$, max $r = 0.085$), indicating acceptable local independence.

5.3 Per-Benchmark Analysis

GSM8K shows the largest raw inflation (14.6 pp) and the highest residual after calibration (2.8 pp), reflecting the extreme sensitivity of free-form math benchmarks to memorization. MMLU (1.1 pp) and ARC-C (1.2 pp) are well calibrated across all dosages.

Per-Benchmark γ_k Estimates. The estimated contamination susceptibility parameters reveal striking heterogeneity:

The γ_k heterogeneity is statistically significant (χ^2 test, $p = 0.002$), confirming that per-benchmark fitting is essential. A joint model with a shared global $\gamma = 0.41$ yields MAE of 2.6 pp, worse than per-benchmark fitting (1.8 pp): the global model overfits MMLU while underfitting GSM8K.

Per-Scale MAE. CONTAMCAL achieves 2.3 pp at 0.5B, 1.4 pp at 1.5B, and 1.6 pp at 3B. The higher MAE at 0.5B reflects degraded signal quality (only S1 effective), leading to noisier exposure estimates. The measurement error model compensates with wider CIs, providing honest uncertainty quantification even in this degraded regime.

5.4 Dose-Response: 3B GSM8K

Table 8 presents the most dramatic calibration case.

At full contamination (d100), the observed score of 69.7% is 57.7 pp above the true score of 12.0%. CONTAMCAL reduces this to a 4.2 pp residual. Even at moderate contamination (d025), naive scores overestimate by $\sim$23 pp, while our calibration achieves a 2.5 pp residual.

Table 8: Dose-response for 3B GSM8K. Naive scores inflate by up to 57.7 pp; CONTAMCAL recovers true performance within 4.2 pp.

Dose	Obs.	Calib.	True	Resid.
d000 (0%)	12.0%	—	12.0%	—
d005 (5%)	20.8%	13.2%	12.0%	+1.2 pp
d025 (25%)	∼35.4%	14.5%	12.0%	+2.5 pp
d050 (50%)	∼51.8%	15.1%	12.0%	+3.1 pp
d100 (100%)	69.7%	16.2%	12.0%	+4.2 pp

Table 9: γ recovery from the simulation sweep. AUC ≥ 0.70 at $N \geq 50$ is the practical identifiability sweet spot (CI $<$ 7 pp, coverage $\geq$ 95%).

AUC	N	Bias	RMSE	CI_{95} (pp)	Cov.
0.55	20	+0.020	0.040	13.3	80%
0.55	50	+0.015	0.026	8.7	90%
0.55	100	−0.003	0.004	6.5	100%
0.65	20	+0.016	0.035	11.8	85%
0.65	50	+0.010	0.020	7.6	93%
0.65	100	−0.004	0.008	5.8	97%
0.70	50	+0.008	0.018	7.0	95%
0.70	100	−0.002	0.006	5.3	98%
0.80	50	+0.006	0.014	6.2	96%
0.80	100	+0.003	0.005	4.8	98%
0.90	50	+0.004	0.010	5.4	97%
0.90	100	+0.002	0.004	4.2	99%

5.5 Identifiability and Simulation Study

We conduct a 120-condition simulation sweep varying signal AUC ($\in \{0.55, 0.65, 0.70, 0.80, 0.90\}$) and number of models ($N \in \{20, 50, 100\}$) at $\gamma_{\text{true}} = 0.5$.

Fisher Information analysis yields AUC ≥ 0.52 as the structural identifiability boundary and AUC ≥ 0.70 as the practical boundary. The latter corresponds precisely to the Min-K++ AUC at 1.5B (∼0.74), confirming that 1.5B is the minimum scale for reliable multi-signal calibration. CI width scales as approximately $1/\sqrt{N}$ (empirical ratio $N_{20}/N_{100} = 2.04$ at AUC=0.55; theoretical: 2.24). Coverage at $N = 20$ falls below nominal (80–85%) due to a known small-N IRT limitation. All fits converge with zero divergences and $\hat{R}_{\max} = 1.012$.

Additional findings: continuous exposure scores produce ∼8% narrower CIs than binary indicators. Informative priors (SD=0.5) narrow CIs by 18%, while vague priors (SD=2.0) widen them by 28%.

5.6 Measurement Error Model

At low noise ($\sigma_e = 0.1$), both models perform similarly. At $\sigma_e = 0.3$ (typical for real signal pipelines based on observed FP/FN rates), the ME model maintains 93% coverage while the simple model drops to 78%. At $\sigma_e = 0.5$, the simple model's coverage collapses to 65%—dangerously miscalibrated—while the ME model maintains 91%. The wider CIs under the ME model reflect genuine epistemic uncertainty: models with degraded signals receive appropriately wider intervals.

Post-hoc noise injection experiments confirm super-linear CI widening: $1.0\times$ noise widens γ CI by 28%, $1.5\times$ by 52%, and $2.0\times$ by 83%.

Table 10: Measurement error (ME) model vs. simple model. At high noise ($\sigma_e = 0.5$), the simple model's coverage collapses to 65%; the ME model maintains 91%.

σ_e	ME Cov.	ME CI (pp)	Simple Cov.	Simple CI (pp)
0.1	95%	5.2	91%	4.8
0.3	93%	7.8	78%	5.1
0.5	91%	11.4	65%	5.3

Table 11: Qwen2.5-7B dose-response and signal quality.

Dosage	MMLU	GSM8K	ARC-C
d000	71.8%	79.2%	87.8%
d005	72.6% (+0.8)	80.4% (+1.2)	88.1% (+0.3)
d025	74.3% (+2.5)	85.0% (+5.8)	89.2% (+1.4)
d100	75.9% (+4.1)	91.7% (+12.5)	90.6% (+2.8)

5.7 Scale Bridging: 7B Validation

To address scale concerns, we extend controlled contamination injection to Qwen2.5-7B.

At 7B, CONTAMCAL achieves 1.4 pp MAE with 95% coverage and Kendall's $\tau = 0.97$—the best performance across all scales. Signal quality continues to improve (S2 AUC: 0.871; S4 AUC: 0.812), confirming that larger models are easier to calibrate.

5.8 Multi-Family Validation: Llama-3.2-1B

To test cross-family generalization, we repeat 3-dosage (d000/d025/d100) controlled SFT injection on Llama-3.2-1B.

CONTAMCAL achieves 2.1 pp MAE with 92% coverage on Llama. The γ estimates are highly consistent across families (Pearson $r = 0.91$), suggesting that contamination susceptibility is primarily a benchmark property (item structure, format, memorability) rather than a model-family artifact. Llama's S2 AUC (0.698) falls just below the 0.70 practical threshold at the $\sim$1B scale, consistent with the simulation sweep predictions.

5.9 Closed-Weight Mode

For closed-weight models lacking logit/embedding access, CONTAMCAL operates in a 2-signal mode using only S3 (ConStat) and S4 (Self-Critique).

At 0.5B, 2-signal mode nearly fails (6.8 pp MAE) because both S3 and S4 are ineffective at this scale. At 1.5B+, 2-signal mode provides meaningful calibration (2.1–2.8 pp MAE), matching DCR's 3.9 pp aggregate while additionally providing credible intervals (86% coverage). Practical guidance: closed-weight mode requires $\geq$1.5B models for reliable calibration.

5.10 Observational Case Studies

RL-Trained Models. We apply CONTAMCAL to Qwen2.5-Instruct (0.5B–7B) and DeepSeek-R1-Distill (1.5B, 7B) without ground truth.

Calibration adjustments are small (0.4–1.6 pp) but CIs are notably wider (6.8 pp average for Instruct, 8.2 pp for DeepSeek) than in the controlled setting (4.2 pp)—reflecting honest uncertainty when signals degrade under RL post-training. GSM8K contamination persists after instruction tuning ($\gamma = 0.42$, CI [0.18, 0.68]), suggesting RLHF does not fully eliminate memorization. Ranking changes are minimal (1 out of 6), but the CI widths provide valuable uncertainty awareness even when rankings are preserved.

Table 12: Signal AUC at 7B (d005). All four signals are effective, with continued improvement over 3B.

Signal	MMLU	GSM8K	ARC-C	Mean
S1 Embedding	0.948	0.971	0.943	0.954
S2 Min-K++	0.851	0.896	0.865	0.871
S3 ConStat	0.872	0.908	0.881	0.887
S4 Self-Crit.	0.795	0.834	0.808	0.812

Table 13: Cross-family γ comparison. High consistency (Pearson $r = 0.91$) suggests contamination susceptibility is a benchmark property, not model-family-specific.

Bench.	Qwen2.5 γ	Llama γ	Diff.
MMLU	0.22	0.19	-0.03
GSM8K	0.85	0.72	-0.13
ARC-C	0.18	0.16	-0.02

Specificity Check. On a known-clean benchmark, CONTAMCAL estimates $\gamma = 0.02$ with CI including zero, confirming the method does not hallucinate contamination where none exists.

Frontier Scale: Qwen2.5-72B. We apply inference-only CONTAMCAL to Qwen2.5-72B.

Signal quality reaches its highest levels (S2 AUC: 0.89; S4 AUC: 0.84), further supporting the scale-dependent trend. GSM8K's γ is significantly nonzero (CI [0.01, 0.37]), indicating detectable contamination even at frontier scale. MMLU and ARC-C γ CIs include zero, consistent with either low contamination or the model's genuine knowledge overshadowing memorization effects. CI widths average 5.5 pp, reasonable for a single-model observational analysis.

6 Discussion

Why Per-Benchmark γ_k? GSM8K ($\gamma = 0.85$) is $\sim 4\times$ more susceptible than MMLU ($\gamma = 0.22$) or ARC-C ($\gamma = 0.18$). Free-form math benchmarks are highly vulnerable because answer memorization directly accelerates generation; multiple-choice benchmarks are partially protected by forced-choice format. This cross-family-consistent pattern (Qwen–Llama Pearson $r = 0.91$) suggests contamination susceptibility is an intrinsic benchmark property.

Honest Uncertainty Quantification. A distinctive feature of CONTAMCAL is that calibration quality degrades gracefully: models with poor signals get wider CIs (0.5B: 2.3 pp MAE, wider CI), while those with strong signals get tighter CIs (7B: 1.4 pp MAE, tighter CI). This is a feature, not a bug—the measurement error model ensures that uncertainty reflects genuine epistemic limitations.

Practical Deployment. Full 4-signal mode requires open-weight models with logit access. For closed-weight models ($\geq$1.5B), the 2-signal mode (S3 + S4 only) achieves 2.1–2.8 pp MAE, matching or exceeding existing methods while additionally providing credible intervals. Each per-benchmark MCMC fit requires $\sim$45 minutes on a single GPU.

7 Limitations

1. **Model family breadth.** Controlled experiments use primarily Qwen2.5, with Llama-3.2-1B as cross-family validation. Broader family coverage (Pythia, Mistral, GPT-family) remains future work.

Table 14: 4-signal vs. 2-signal (closed-weight) mode. At 1.5B+, 2-signal mode matches or outperforms the best existing baseline (DCR: 3.9 pp).

Scale	4-Sig MAE	2-Sig MAE	Gap	2-Sig AUC
0.5B	2.3 pp	6.8 pp	+4.5 pp	0.508
1.5B	1.4 pp	2.8 pp	+1.4 pp	0.782
3B	1.6 pp	2.1 pp	+0.5 pp	0.831
Avg.	1.8 pp	3.9 pp	+2.1 pp	—

Table 15: Observational calibration on RL-trained models (MMLU).

Model	Raw	Calib.	95% CI	ΔRank
Qwen-0.5B-Inst	47.2%	46.8%	[43.1, 50.5]	0
Qwen-1.5B-Inst	61.5%	60.2%	[56.8, 63.6]	0
Qwen-3B-Inst	68.3%	67.1%	[63.5, 70.7]	0
Qwen-7B-Inst	74.8%	73.2%	[70.1, 76.3]	−1
DS-R1-1.5B	58.9%	58.4%	[53.8, 63.0]	+1
DS-R1-7B	75.2%	74.6%	[71.0, 78.2]	0

2. **Unidimensional θ.** We use a single ability parameter per model-benchmark, which is a simplification for multifaceted benchmarks like MMLU (Reckase, 2009). Multidimensional IRT is a natural extension.

3. **SFT $\neq$ real contamination.** LoRA SFT injection is a controlled proxy for real-world contamination (which may occur during pre-training, RLHF, or data mixture). Observational studies provide external validity but lack ground truth.

4. **Local independence.** With a single model producing all item responses, strict local independence is violated. Our per-benchmark fitting and residual correlation analysis mitigate but do not eliminate this concern.

5. **Closed-weight degradation.** The 2-signal mode fails at 0.5B and provides degraded but usable calibration at 1.5B+.

6. **Computational cost.** Per-benchmark MCMC requires $\sim$45 minutes; signal computation adds overhead for each model evaluated.

8 Conclusion

We introduced CONTAMCAL, a framework that bridges contamination detection and score calibration via contamination-augmented 4PL IRT. On controlled experiments across 0.5B–7B models and two model families, CONTAMCAL recovers true benchmark performance within 1.8 pp MAE with 94% credible interval coverage, while providing honest uncertainty quantification that widens intervals when signals are unreliable. Our systematic signal characterization reveals scale-dependent effectiveness with a practical threshold at 1.5B, and identifies Min-K++ directional reversal as a cautionary finding for the detection community. Fisher Information analysis provides theoretical grounding for identifiability bounds. Observational studies on RL-trained models up to 72B demonstrate directional external validity.

CONTAMCAL is a statistical safety net for benchmark evaluation: it does not eliminate contamination, but it measures its effect and subtracts it out with proper uncertainty. Future work includes extending to multidimensional IRT, broader model families, and integration with dynamic benchmark curation. Data attribution methods (Pruthi et al., 2020; Park et al., 2023) offer promising additional contamination signals for future integration.

Table 16: Qwen2.5-72B observational results. Signal quality reaches its highest level; GSM8K γ is significantly nonzero.

Bench.	**Raw**	**Calib.**	**95% CI**	γ_k
MMLU	85.4%	84.2%	[81.5, 86.9]	0.12 [−0.02, 0.28]
GSM8K	90.8%	89.3%	[86.2, 92.4]	0.18 [0.01, 0.37]
ARC-C	94.3%	93.8%	[91.4, 96.2]	0.09 [−0.04, 0.23]

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